

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 13 · LECTURE NOTES

Claiming a Biological Result

What the analysis established, and what it did not.

MODULE OUTCOMES

- State the inference chain behind a typical claim.
- Identify the standard overclaims.
- Explain why a negative control is required.
- Place this course in the semester and the program.
- Write an honest methods and claims section.

1 The chain

A COMPUTATIONAL BIOLOGY RESULT IS A STATISTICAL STATEMENT ABOUT A MODEL APPLIED TO A PARTICULAR COHORT'S MEASUREMENTS

The measurement — what the instrument actually emitted, with its known biases: counts or intensities, not biological quantities (Module 01 §2).

The processing — alignment tool and parameters, reference build, annotation release, normalisation, and filtering — each of which has defensible alternatives that change the result (Modules 04, 05, 10, and 12).

The model and the test — including the number of tests performed and the correction applied (Module 10 §3), and the effect sizes alongside the p-values.

And the population — who was measured, and whether the batch and ancestry structure was examined or merely left unexamined (Module 10 §4, Module 11 §2). **Five things, and a typical abstract names none of them**, which is why the methods section is where the content is.

The claim	The correction
'We identified the gene for X.'	A variant in a region statistically associated with X, in this cohort — see the note (Module 11 §3).
'Gene Y is upregulated in disease.'	Reads assigned to an annotation and counted — normalised how, tested how, among how many tests? (Module 10 §§1 and 3.)
'This protein functions as Z.'	Predicted structure, inferred by homology, or measured activity? Three very different claims (Module 09 §3).

The claim	The correction
'BLAST found no homologue, so there is none.'	These parameters, this database version, this seed length (Module 04 §1).
'The tree shows that A and B are sister taxa.'	Under this evolutionary model, from this alignment, with this support value (Module 06 §4).
'The results are reproducible.'	Same data and same code, or new data? Those are different standards (Module 12 §4).

'The gene for X' is the one to refuse hardest — it asserts causality, specificity, and mechanism all at once, and an association study establishes none of them (Module 11 §4's closing chain). Each correction above names a step the claim skipped, which is why the corrections are specific rather than general scepticism: there is always an identifiable missing clause.

2 The negative control

RUN THE ANALYSIS ON DATA WHERE THE ANSWER SHOULD BE NOTHING, AND REPORT WHAT IT GIVES

Permute the condition labels, shuffle the sequence, or use a comparison that should show no difference — and run the entire pipeline on it, not a simplified version, because the point is to exercise the parts you are not thinking about.

Because it tests the whole chain including those parts — a label leak, a filter applied after the train-test split, an index built on the wrong file, a join on the wrong key — all of which produce signal that no amount of reading the code reliably catches (CSCE 633 Module 12's leakage material, in a different setting).

And a pipeline that finds two hundred significant genes in permuted labels has told you something decisive — which no amount of careful reasoning would have, and which takes one rerun to learn.

Which is Module 03 §2's empirical null, generalised from a score to an entire pipeline — and it is why Project 2 fails without one, whatever else it found, since an uncontrolled finding is not assessable.

3 The honest write-up

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THE DATA
  source, accession number, and what was excluded
  and why

THE PIPELINE
  tool versions, reference build AND annotation
  release, and every parameter that differs from
  the defaults

THE STATISTICS
  the model, the number of tests INCLUDING the
  ones you ran and discarded, the correction, and
  the effect sizes alongside the p-values

THE CONTROLS
  the negative control and its result
  any positive control, and whether it was
  recovered

THE STRUCTURE
  batch and ancestry: examined, or explicitly
  stated as unexaminable from the available
  metadata

AND THE ARTEFACT ACCOUNT
  "if this were batch structure it would look like
  X, and here is whether it does"
  which is the sentence that distinguishes an
  analysis from a result.

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The artefact account is what distinguishes an analysis from a result — say what the finding would look like if it were not real, and then check. It is the difference between presenting a number and having interrogated it, it is cheap, and it is the single most useful paragraph in a methods section. **Project 2 requires it**, and it is the analytical skill the whole course is aimed at.

4 The semester and the program

Course	The external model	What it costs you
CSCE 640	Physics.	One measurement per run, and interference is the only resource — the gate set and the noise are given.
CSCE 628 (this one)	Biology.	Noisy data, and no ground truth to check the answer against — so validation is by proxy and controls.
CSCE 717	Strategic agents.	Inputs chosen to benefit whoever sent them — so correctness must include the incentive to report honestly.

In all three courses the rules come from outside computer science and are not negotiable — and the first job in each is to state them accurately rather than to assume them away. That is a different discipline from the earlier semesters, where the machine, the data, and the inputs were all under your control. **The model-from-outside framing is Semester 12's result**, and this course's version of it is that the data was generated by an experiment you did not run and cannot repeat.

WHERE THIS COURSE LEAVES YOU

You can derive and implement the core algorithms — alignment, Viterbi, and RNA folding — and, more usefully, recognise that they are three variations on one recurrence over different subproblem structures (Module 02 §4's closing point).

You know where the scores come from and what a null model is for (Module 03) — **which is the habit that transfers furthest beyond this subject**, and is the same move as CSCE 658's permutation tests and CSCE 676's significance material.

And you know the inference failures that actually sink findings — uncorrected multiple testing, confounded batches, population structure, and irreproducible pipelines — which is precisely the part a pure algorithms course omits and the part that determines whether any of the algorithms were worth running.

The closing rule is the program's, unchanged across thirty-five courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it is the five-part methods section and the artefact account — because 'we found a gene' is a claim about a pipeline, and the pipeline is where the claim can be checked.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Ioannidis — Why most published research findings are false (free)	§1 — the argument, which is about multiple testing and prior probability and applies directly.
Begley & Ioannidis — Reproducibility in science (free)	§2 — what failed to replicate and what the successful analyses had in common.
The ARRIVE and MIAME-style reporting guidelines (free)	§3 — structured reporting checklists, which are a good basis for your own methods section.
Leek & Peng — Statistics: P values are just the tip of the iceberg (free)	§§1 and 3 — the pipeline, not the test, is where the inference happens.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take one published abstract** and enumerate its five-part chain from the methods.
- 2 **Find one that does not supply enough to do that.**
- 3 **Correct six overclaims** in your own words.
- 4 **Rewrite one 'the gene for X' headline** accurately.
- 5 **Run a negative control** on an analysis you have already done.
- 6 **Deliberately introduce a leak** and confirm the control catches it.
- 7 **Write the six-part methods section** for your Project 2.
- 8 **Write the artefact account**, specifically.
- 9 **State each Semester 12 course's external model.**
- 10 **Project 2 is now due.** Submit the reproducible pipeline, the multiple-testing accounting, the negative control and its result, the batch and confounding examination, and the honest claim with its artefact account.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the five parts of the inference chain.
- 2 Why does a typical abstract not supply them?
- 3 Correct six standard overclaims.
- 4 Why is 'the gene for X' the worst of them?
- 5 What does a negative control test that reasoning does not?
- 6 Why must it run the whole pipeline?
- 7 Give the six parts of an honest write-up.
- 8 What is the artefact account, and why does it matter?
- 9 State each Semester 12 course's external model and its cost.
- 10 State the closing rule in this subject's terms.